Machine Learning Techniques for Reconstructing the Assembly History of the Milky Way

by

Andrea Sante

A thesis submitted in partial fulfilment of the requirements of
Liverpool John Moores University for the degree of
Doctor of Philosophy

Month Year

Declaration

The work presented in this thesis was carried out at the Astrophysics Research Institute, Liverpool John Moores University. Unless otherwise stated, it is the original work of the author.

While registered as a candidate for the degree of Doctor of Philosophy, for which submission is now made, the author has not been registered as a candidate for any other award. This thesis has not been submitted in whole, or in part, for any other degree.

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Abstract

[Abstract goes here.]

In this thesis I develop data-driven and machine-learning methodologies that improve upon those traditionally adopted for the reconstruction of the Milky Way's assembly history. Throughout, I exploit cosmological hydrodynamical zoom-in simulations of Milky Way-mass haloes—in which the origin of every star particle is known—to build, validate and calibrate each method before applying it to observations. The work follows the logical arc of the standard workflow and extends it end to end: classifying accreted versus in-situ stars (Chapter 2), grouping accreted debris into progenitors (Chapter 3), characterising the properties of those progenitors at accretion (Chapter 4), and finally applying this machinery to the real Galaxy (Chapter 5).

Publications

During the course of the preparation of this thesis, the work within Chapters 2, 3, 4, and 5 has been presented in the following jointly authored publications:

1 *'Applying machine learning to Galactic Archaeology: how well can we recover the origin of stars in Milky Way-like galaxies?'*
Sante, A., Font, A. S., Ortega-Martorell, S., Olier, I., & McCarthy, I. G.

2 *'Optimized HDBSCAN clustering for reconstructing the merger history of the Milky Way: applications and limitations'*
Sante, A., Font, A. S., Mistry, I., Ortega-Martorell, S., & Olier, I.

3 *'GalactiKit: reconstructing mergers from $z = 0$ debris using simulation-based inference in Auriga'*
Sante, A., Kawata, D., Font, A. S., & Grand, R. J. J.

4 *'A simulation-based inference of the Milky Way merger history'*
Sante, A., Font, A. S., Kawata, D., Makinen, T. L., & Grand, R. J. J.

Publication 1 forms the basis of Chapter 2. It presents several machine learning models trained on the `ARTEMIS` cosmological zoom-in simulations to separate accreted from in-situ stars in Milky Way-mass galaxies. All analysis was carried out by A. Sante with major scientific contributions from A.S. Font. The implementation of the machine learning models was reviewed and supported by

Publication 2 forms the basis of Chapter 3. It investigates the effectiveness of the HDBSCAN clustering algorithm for recovering the progenitors of Milky Way-type galaxies using the Auriga simulations, introducing a 12-dimensional chemodynamical feature space and an optimisation strategy to reduce fragmentation. All analysis was carried out by A. Sante.

Publication 3 forms the basis of Chapter 4. It presents GalactiKit, a simulation-based inference framework for estimating the infall time, stellar mass, halo mass and mass ratio of disrupted progenitors from the properties of their $z = 0$ debris in the Auriga simulations. All analysis was carried out by A. Sante.

Publication 4 forms the basis of Chapter 5. It applies a normalizing-flow simulation-based inference framework, trained on Auriga debris samples, to a local sample of accreted stars in the Milky Way, inferring the accretion times and masses of several known merger events and placing constraints on the growth of the Galactic halo. All analysis was carried out by A. Sante.

Acknowledgements

[Acknowledgements go here.]

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Appendix A

Merger posterior distributions of MW accreted substructures

This appendix supplements Chapter 5. The posterior distributions of the lookback infall time, stellar and halo mass, and halo mass merger ratio of the MW accreted satellites were estimated from the present-day chemodynamics of their debris using an ensemble of Masked Autoregressive Flows following the GalactiKit methodology. In Chapter 5, samples from these distributions were used to compute summary statistics, providing a quantitative description of the properties of each merger event and, overall, of the MW assembly history.

In this section, the posterior sample distributions for the accreted substructures are reported in individual figures: GES (Fig. A.1), Helmi streams (Fig. A.2), Heracles (Fig. A.3), I'itoi (Fig. A.4), LMS-1/Wukong (Fig. A.5), Sagittarius (Fig. A.6), and Thamnos (Fig. A.10). For Sequoia, we provide three distributions conditioned on debris selected with different criteria: *K19* (Fig. A.7), *M19* (Fig. A.8), and *N20* (Fig. A.9). Each distribution is presented as a corner plot of the inferred galaxy merger properties, with one-dimensional density distributions showing the median and the 16th–84th percentile range. Samples are coloured consistently with the substructure colour scheme adopted in the analysis.

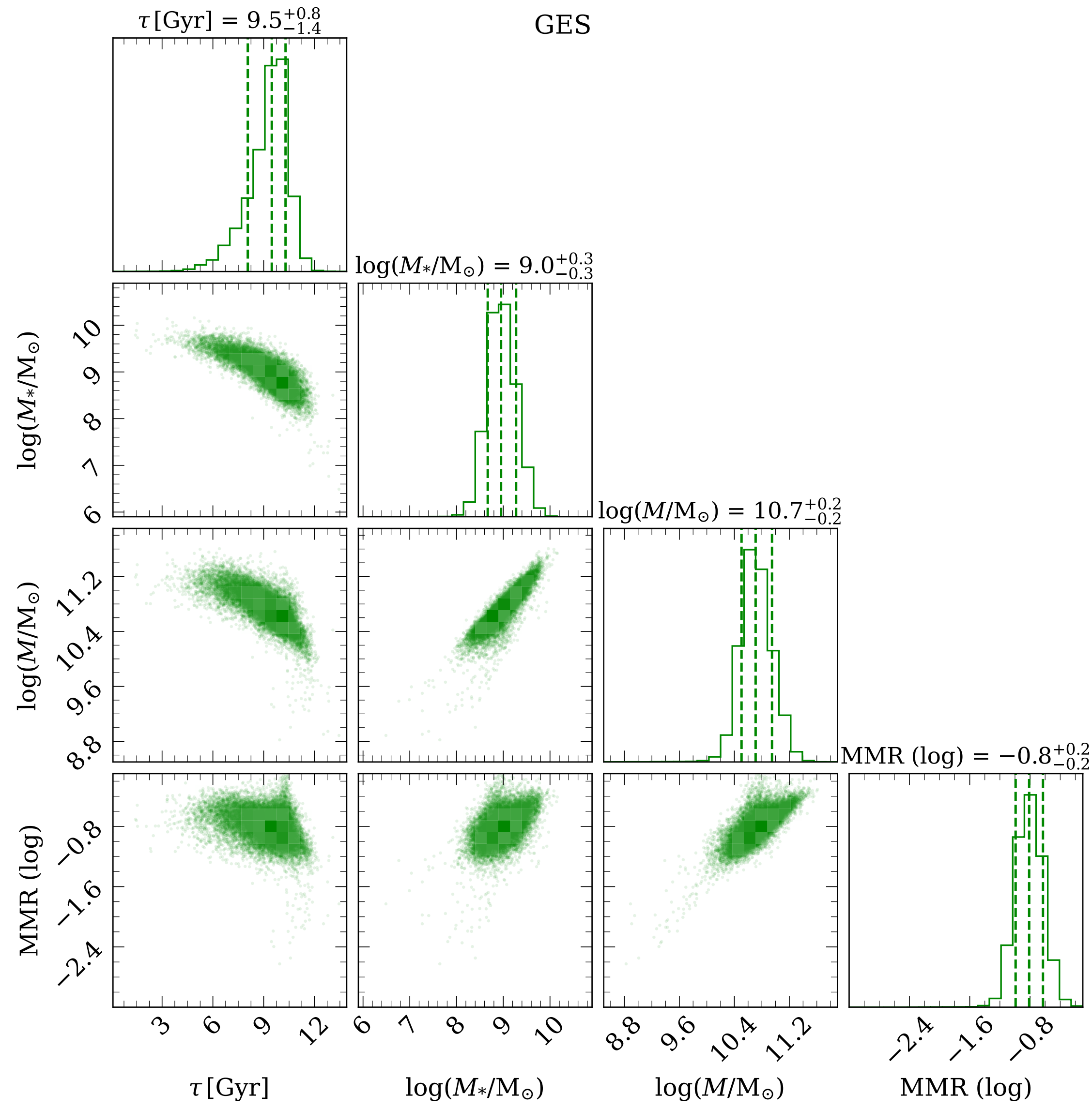

GES
τ [Gyr] = $9.5^{+0.8}_{-1.4}$
log(M_*/M$_\odot$) = $9.0^{+0.3}_{-0.3}$
log(M/M$_\odot$) = $10.7^{+0.2}_{-0.2}$
MMR (log) = $-0.8^{+0.2}_{-0.2}$
log(M_*/M$_\odot$)
log(M/M$_\odot$)
MMR (log)
τ [Gyr]
log(M_*/M$_\odot$)
log(M/M$_\odot$)
MMR (log)

Figure A.1

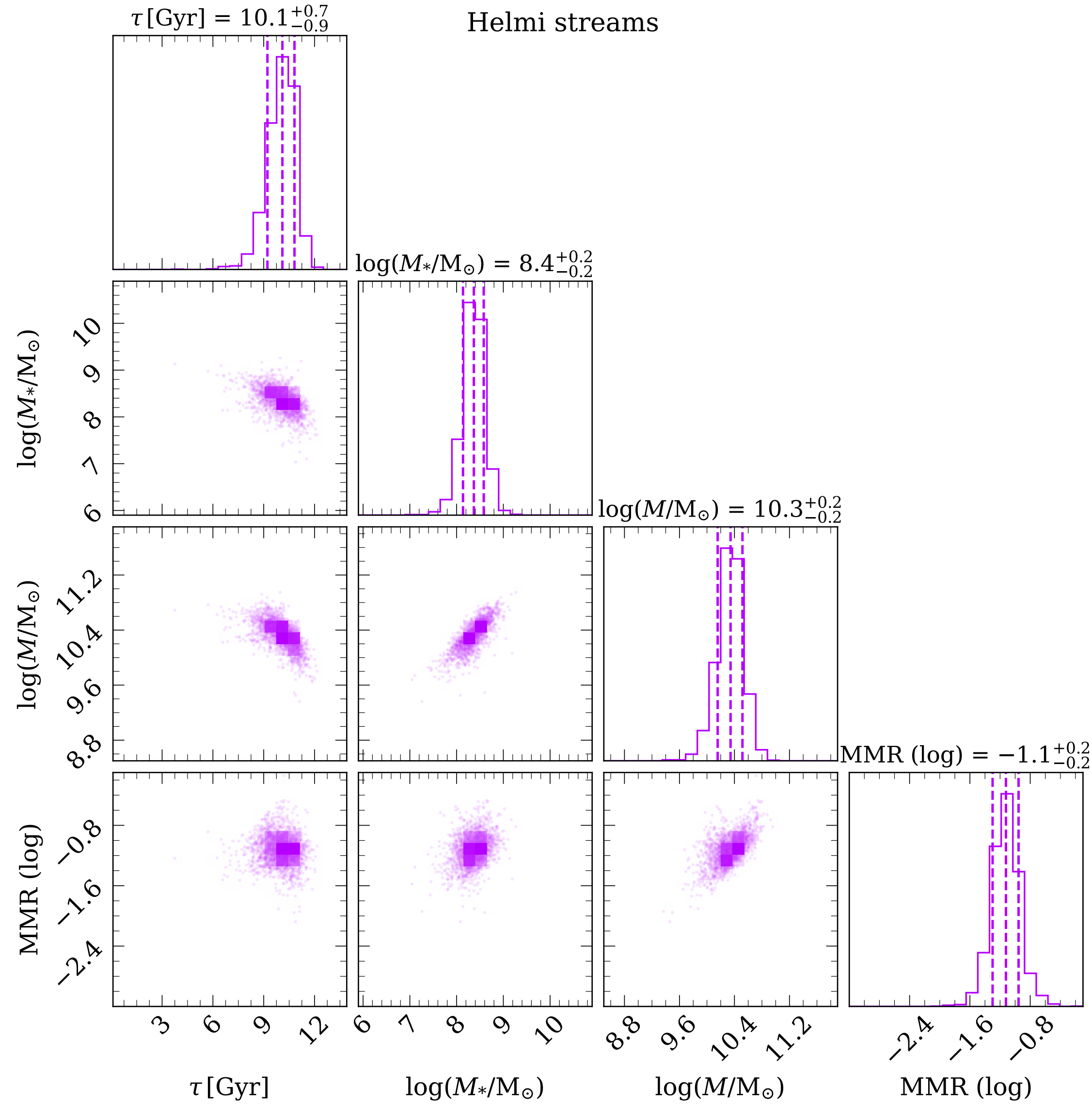
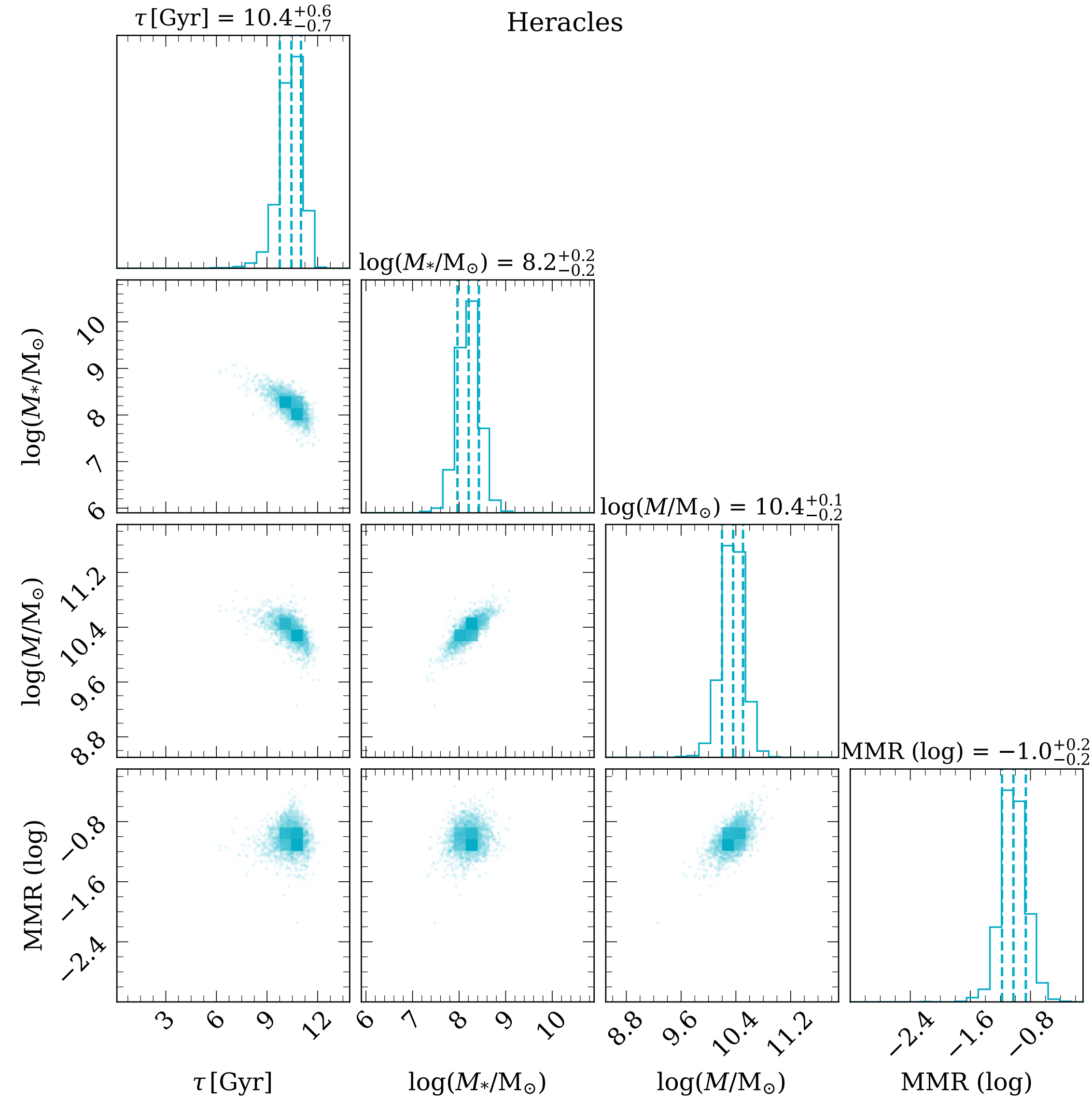
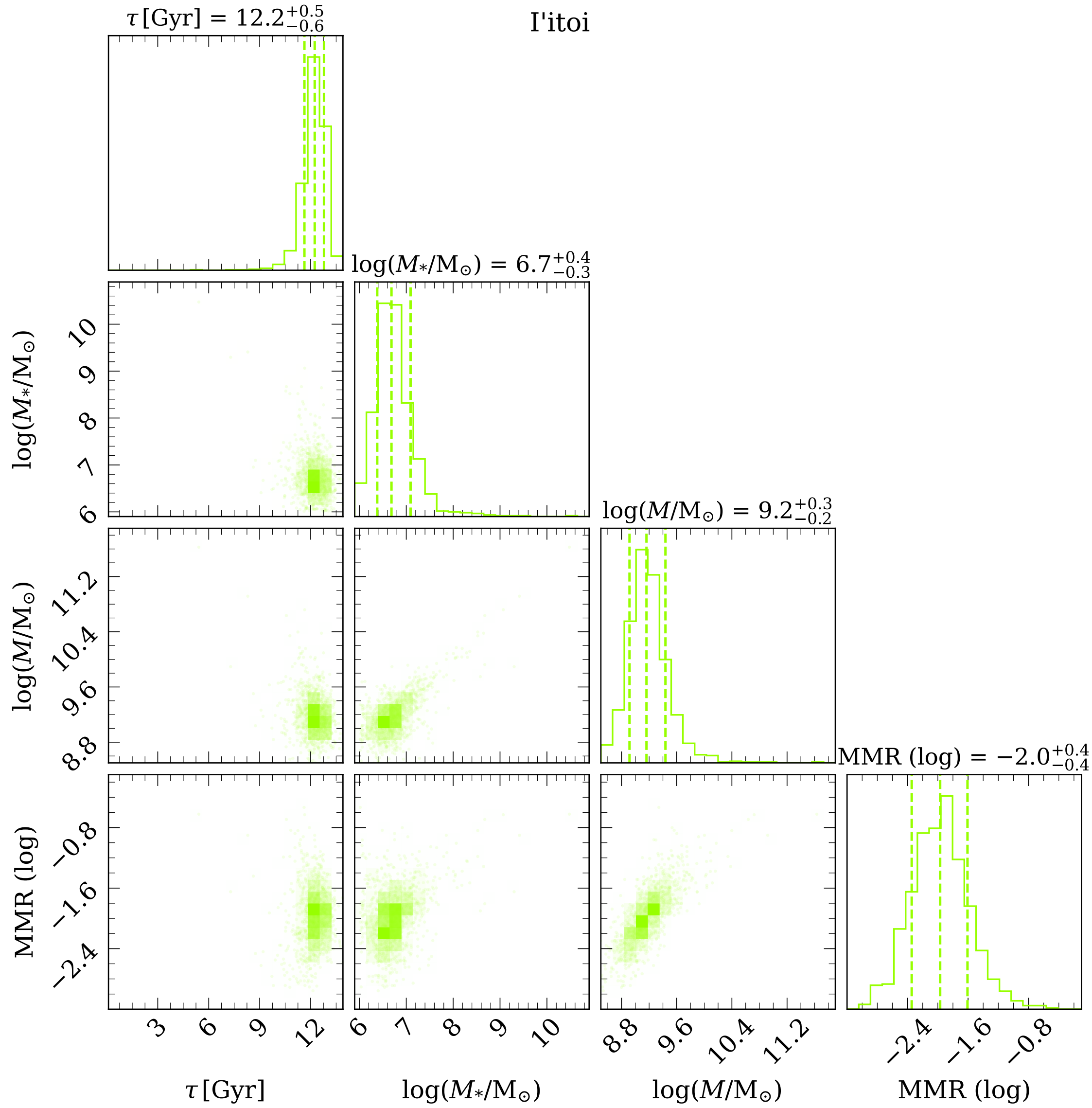

Helmi streams
τ [Gyr] = 10.1$^{+0.7}_{-0.9}$
log(M*/M☉) = 8.4$^{+0.2}_{-0.2}$
log(M/M☉) = 10.3$^{+0.2}_{-0.2}$
MMR (log) = −1.1$^{+0.2}_{-0.2}$
log(M*/M☉)
log(M/M☉)
MMR (log)
τ [Gyr]
log(M*/M☉)
log(M/M☉)
MMR (log)

Figure A.2



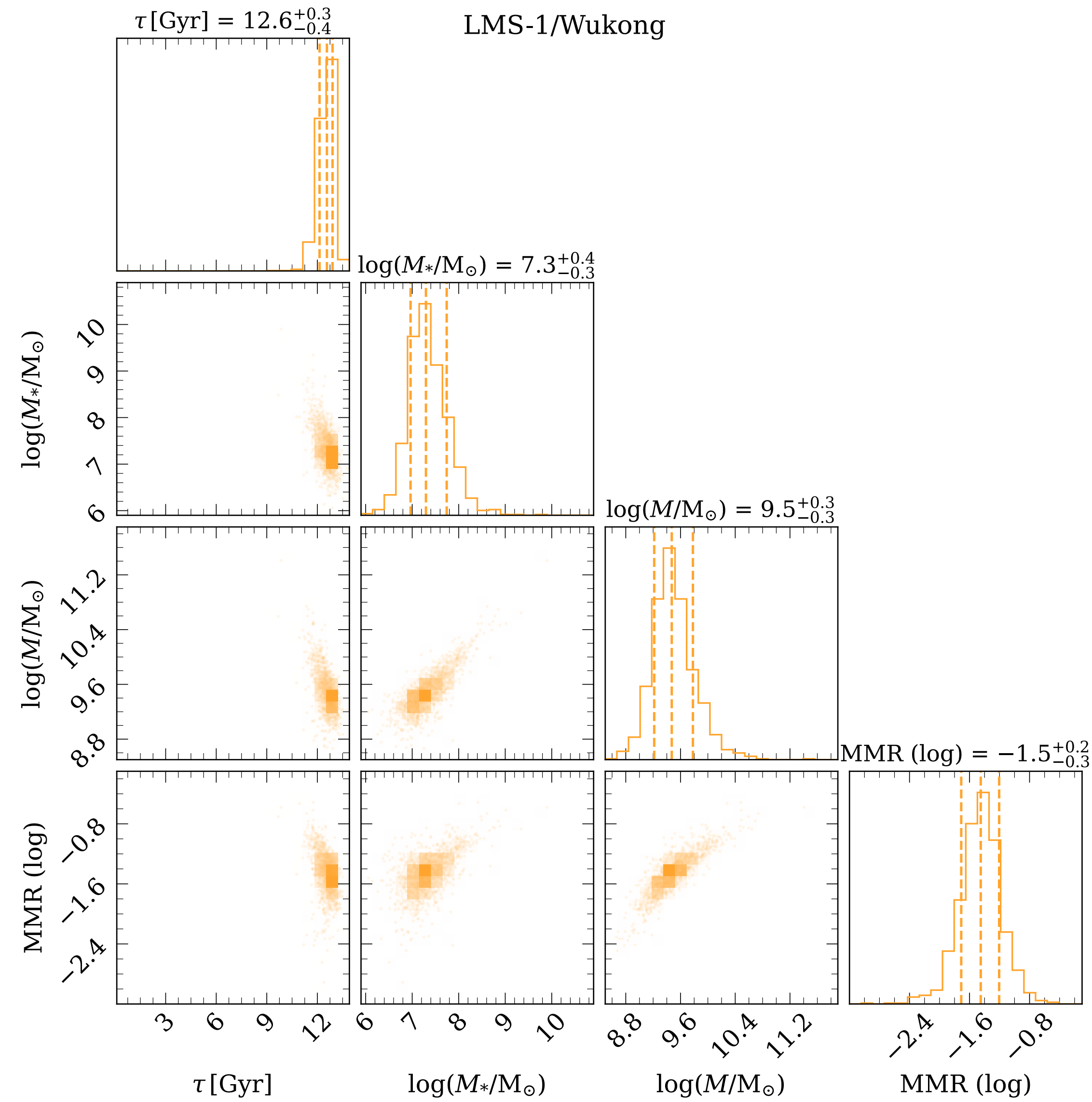
Heracles
$\tau\,[\mathrm{Gyr}] = 10.4^{+0.6}_{-0.7}$
$\log(M_*/\mathrm{M}_\odot) = 8.2^{+0.2}_{-0.2}$
$\log(M/\mathrm{M}_\odot) = 10.4^{+0.1}_{-0.2}$
MMR (log) $= -1.0^{+0.2}_{-0.2}$
$\tau\,[\mathrm{Gyr}]$
$\log(M_*/\mathrm{M}_\odot)$
$\log(M/\mathrm{M}_\odot)$
MMR (log)

Figure A.3



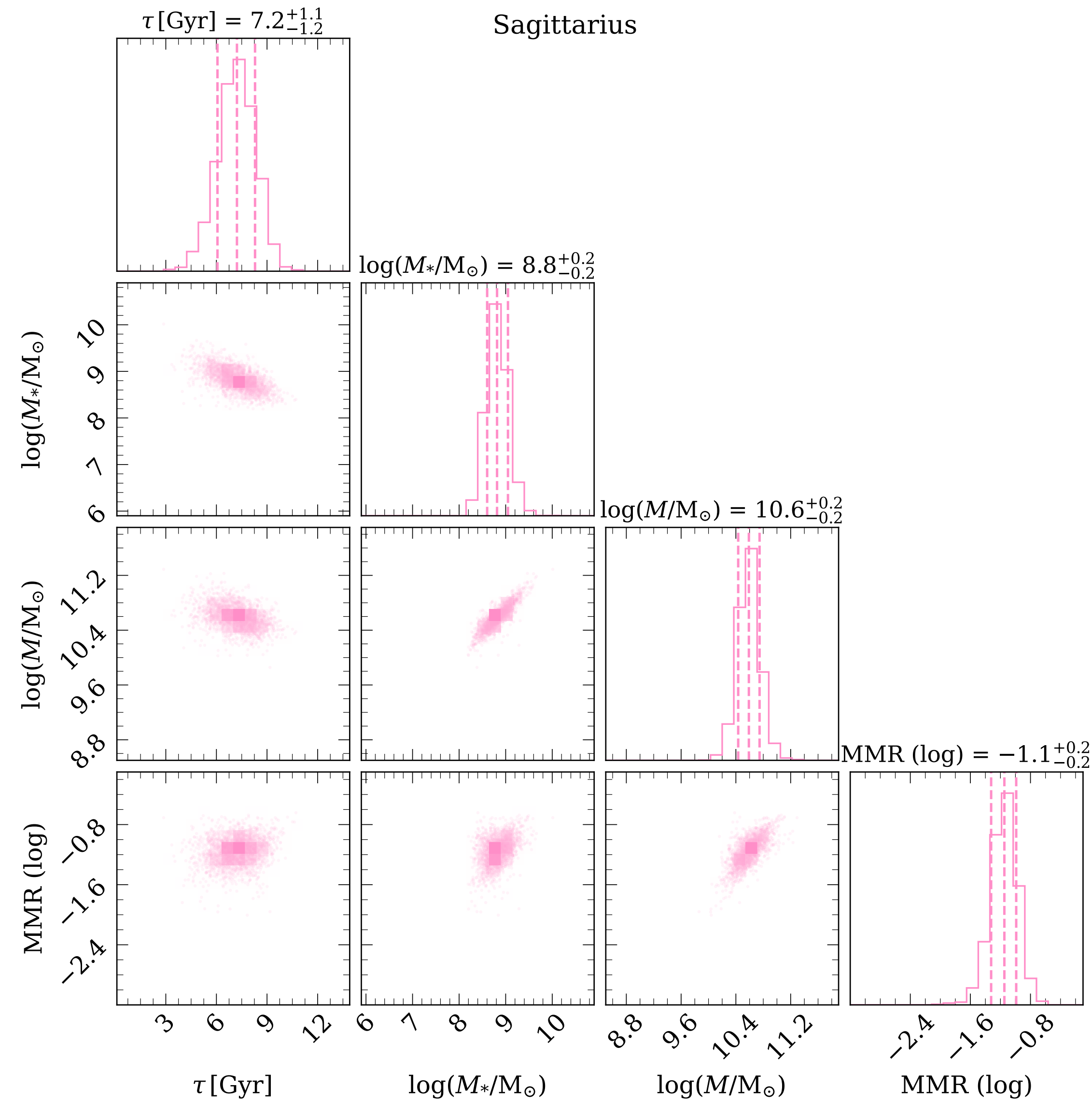
I'itoi
τ [Gyr] = $12.2^{+0.5}_{-0.6}$
log(M_*/M$_\odot$) = $6.7^{+0.4}_{-0.3}$
log(M/M$_\odot$) = $9.2^{+0.3}_{-0.2}$
MMR (log) = $-2.0^{+0.4}_{-0.4}$
τ [Gyr]
log(M_*/M$_\odot$)
log(M/M$_\odot$)
MMR (log)

Figure A.4



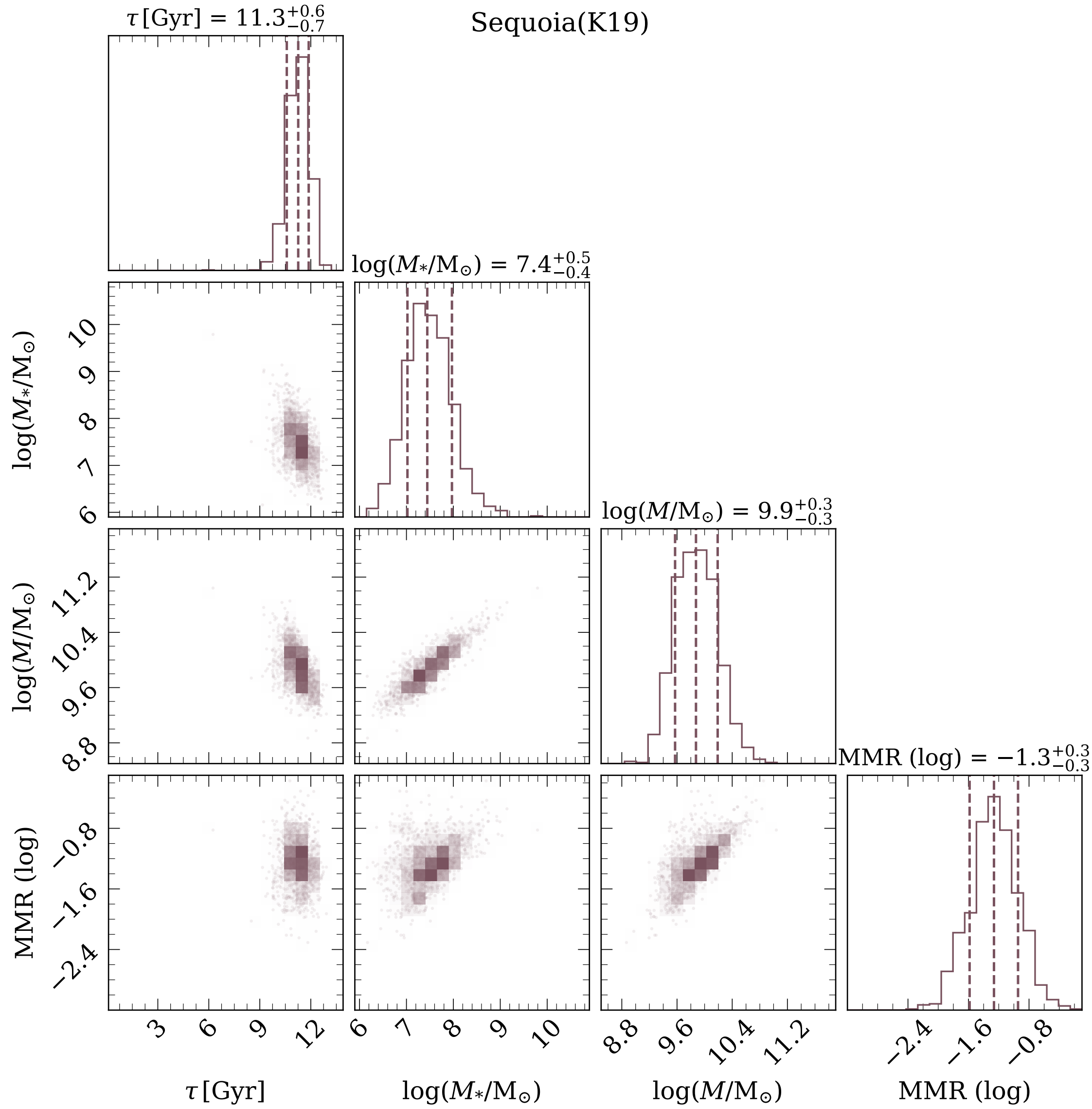
LMS-1/Wukong
τ [Gyr] = $12.6^{+0.3}_{-0.4}$
$\log(M_*/M_\odot) = 7.3^{+0.4}_{-0.3}$
$\log(M/M_\odot) = 9.5^{+0.3}_{-0.3}$
MMR (log) = $-1.5^{+0.2}_{-0.3}$
τ [Gyr]
$\log(M_*/M_\odot)$
$\log(M/M_\odot)$
MMR (log)

Figure A.5



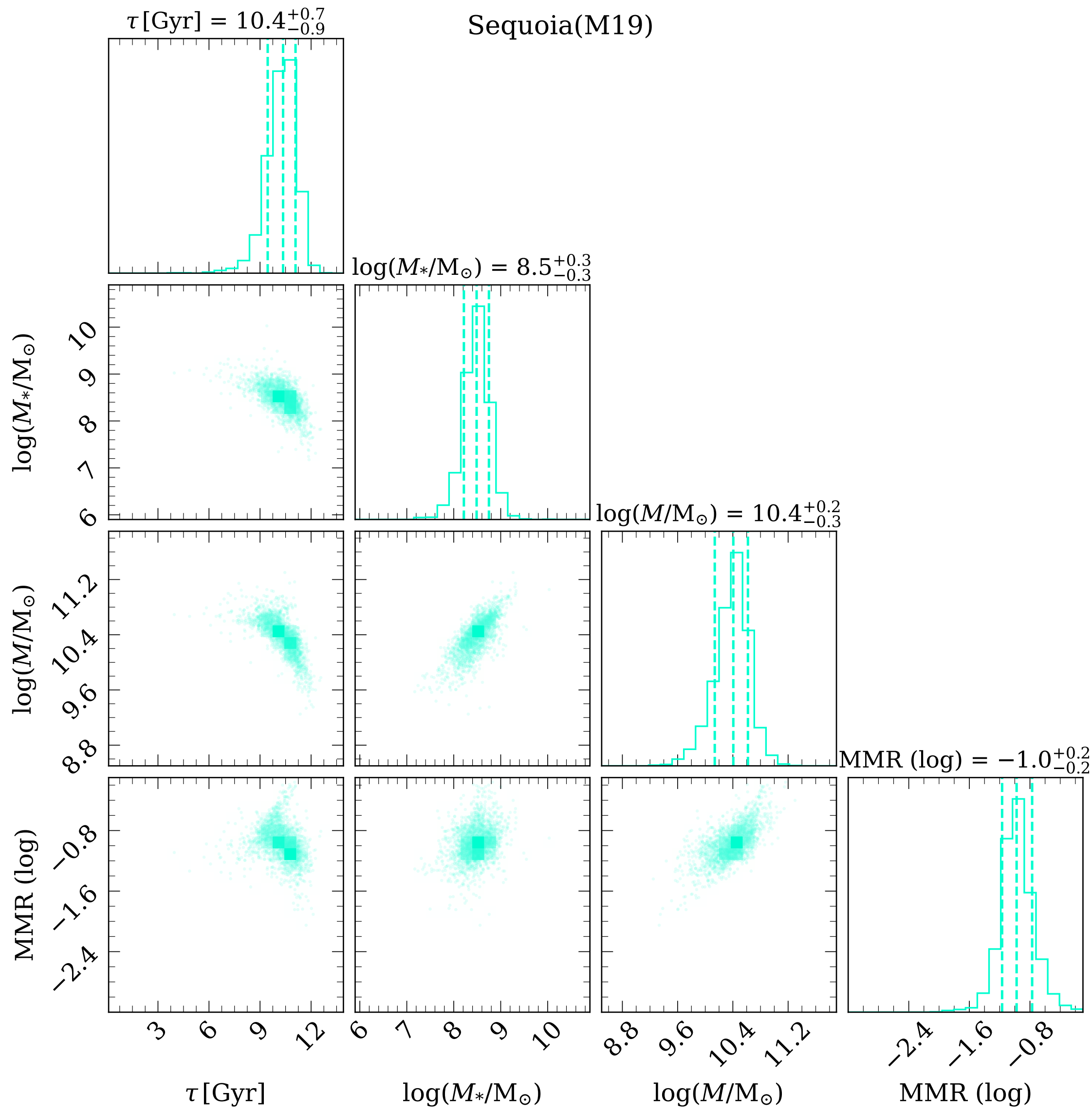
$\tau\,[\mathrm{Gyr}] = 7.2^{+1.1}_{-1.2}$
Sagittarius
$\log(M_*/\mathrm{M}_\odot) = 8.8^{+0.2}_{-0.2}$
$\log(M/\mathrm{M}_\odot) = 10.6^{+0.2}_{-0.2}$
MMR (log) = $-1.1^{+0.2}_{-0.2}$
$\log(M_*/\mathrm{M}_\odot)$
$\log(M/\mathrm{M}_\odot)$
MMR (log)
$\tau\,[\mathrm{Gyr}]$

Figure A.6



Sequoia(K19)
$\tau\,[\mathrm{Gyr}] = 11.3^{+0.6}_{-0.7}$
$\log(M_*/\mathrm{M}_\odot) = 7.4^{+0.5}_{-0.4}$
$\log(M/\mathrm{M}_\odot) = 9.9^{+0.3}_{-0.3}$
$\mathrm{MMR\ (log)} = -1.3^{+0.3}_{-0.3}$
$\tau\,[\mathrm{Gyr}]$
$\log(M_*/\mathrm{M}_\odot)$
$\log(M/\mathrm{M}_\odot)$
MMR (log)

Figure A.7



$\tau\,[\mathrm{Gyr}] = 10.4^{+0.7}_{-0.9}$
Sequoia(M19)
$\log(M_*/\mathrm{M}_\odot) = 8.5^{+0.3}_{-0.3}$
$\log(M/\mathrm{M}_\odot) = 10.4^{+0.2}_{-0.3}$
MMR (log) $= -1.0^{+0.2}_{-0.2}$
$\log(M_*/\mathrm{M}_\odot)$
$\log(M/\mathrm{M}_\odot)$
MMR (log)
$\tau\,[\mathrm{Gyr}]$
$\log(M_*/\mathrm{M}_\odot)$
$\log(M/\mathrm{M}_\odot)$
MMR (log)

Figure A.8

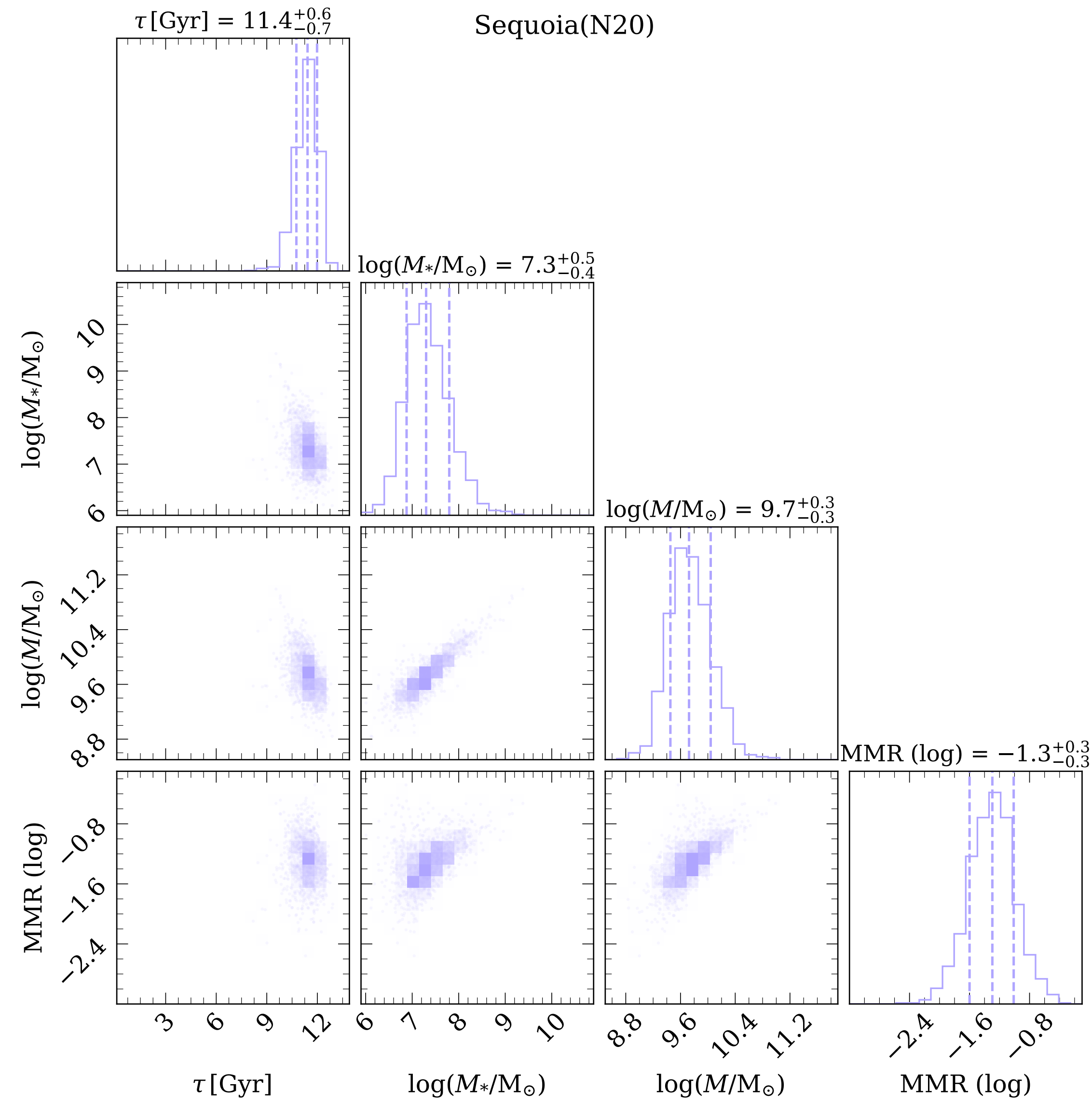

Sequoia(N20)
τ [Gyr] = 11.4$^{+0.6}_{-0.7}$
log(M_*/M$_\odot$) = 7.3$^{+0.5}_{-0.4}$
log(M/M$_\odot$) = 9.7$^{+0.3}_{-0.3}$
MMR (log) = −1.3$^{+0.3}_{-0.3}$
log(M_*/M$_\odot$)
log(M/M$_\odot$)
MMR (log)
τ [Gyr]
log(M_*/M$_\odot$)
log(M/M$_\odot$)
MMR (log)

Figure A.9

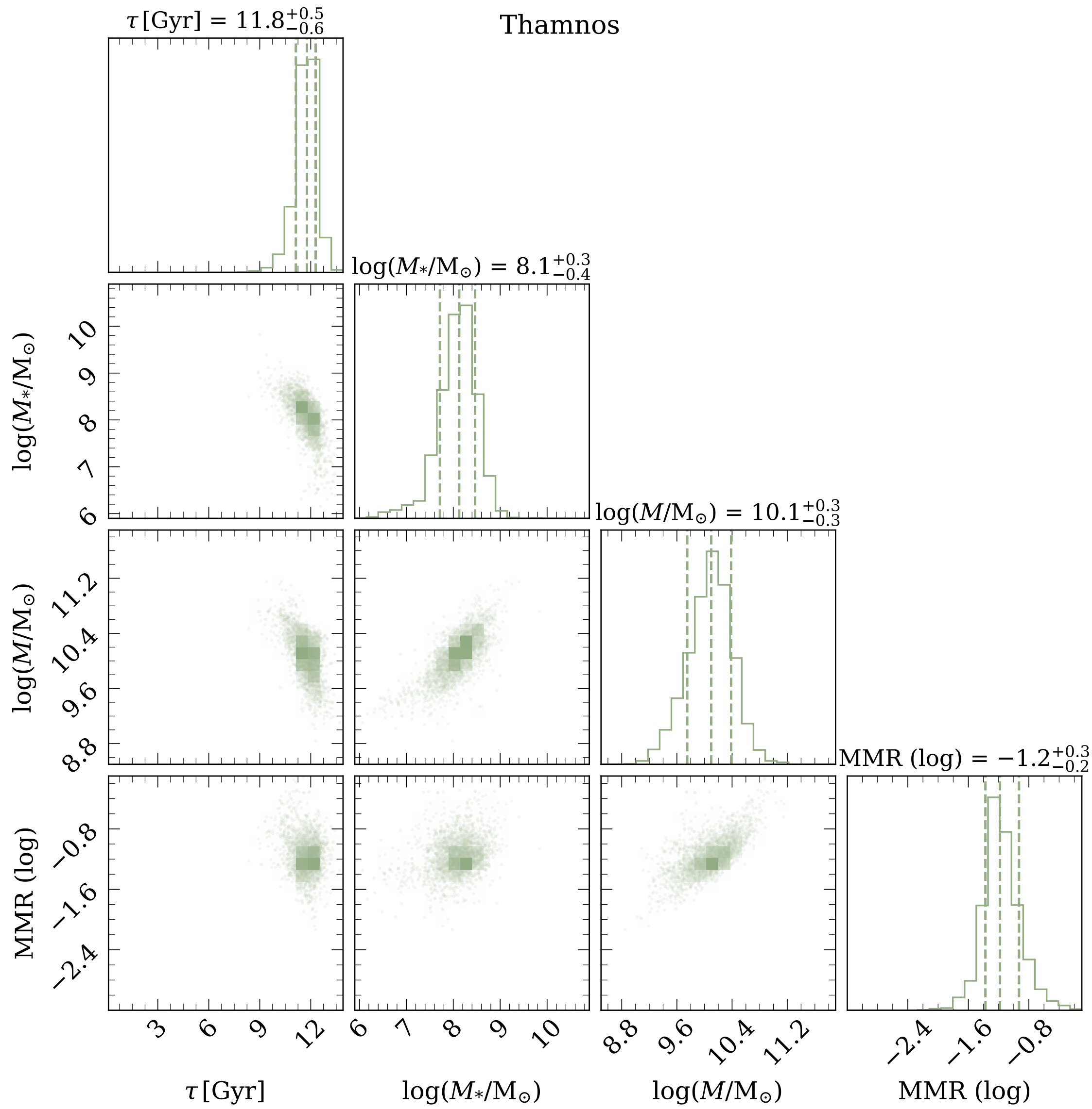

Thamnos
τ [Gyr] = $11.8^{+0.5}_{-0.6}$
$\log(M_*/\mathrm{M}_\odot)$ = $8.1^{+0.3}_{-0.4}$
$\log(M/\mathrm{M}_\odot)$ = $10.1^{+0.3}_{-0.3}$
MMR (log) = $-1.2^{+0.3}_{-0.2}$
τ [Gyr]
$\log(M_*/\mathrm{M}_\odot)$
$\log(M/\mathrm{M}_\odot)$
MMR (log)

Figure A.10

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